	Parul University **Faculty of Engineering and Technology** **Parul Institute of Engineering and Technology** **Department: IT/E**

Name of the teacher: Komal Makwana **Subject Name: Probability, Statistics and Numerical Method (303191251)**	**Hours/Week: 4 Hours**

Sr. No.	Name of Topic	Planned Date
CH 4	**Solution of a System of Linear Equations and Roots of Algebraic and Transcendental Equations**	**7**
1	Gauss-Jacobi method	20/1/26
2	Gauss Seidel method	**20/1/26**
3	Bisection method	21/1/26
4	Regular falsi method	23/1/26
5	Newton Raphson method	23/1/26
6	Derivation by using Newton-Raphson methods	27/1/26
7	Rate of convergence	27/1/26
CH 5	**Finite Differences and Interpolation**	**7**
8	Finite Differences	28/1/26
9	Relation between operators	30/1/26
10	Interpolation- Newton’s forward interpolation	3/2/26
11	Interpolation- Newton’s Backward interpolation	3/2/26
12	Lagrange’s formula for unequal intervals	4/2/26
13	Newton’s divided difference method	4/2/26
14	Newton’s divided difference examples	6/2/26
CH 6	**Numerical Integration and Numerical solution of Ordinary Differential Equations**	**7**
15	Trapezoidal	10/2/26
16	Simpson’s $\frac{1}{3rd}$ formulae	10/2/26
17	Simpson’s $\frac{3}{8th}$ formulae	11/2/26
18	Gaussian quadrature formulae	**11/2/26**

19	Taylor series method	13/2/26
20	Euler method and Euler’s Modified Methods	17/2/26
21	Runge-Kutta method of order two & four	17/2/26
CH 1	**Correlation, Regression and Curve Fitting:**	**11**
22	Correlation and its types	18/2/26
23	Karl-Pearson method for correlation	20/2/26
24	Regression line X on Y	20/2/26
25	Regression line Y on X	24/2/26
26	Rank Correlation (Non-repeated)	24/2/26
27	Fit a straight line	25/2/26
28	Fit a second-degree parabola	27/2/26
29	Fit a second-degree parabola	27/2/26
30	Fit an exponential curve	3/3/26
31	Fit an exponential curve	6/3/26
32	Fit a Power Curve	6/3/26
CH 2	**Probability and Probability distributions**	**13**
33	Probability basic examples	10/3/26
34	Mutually Exclusive events, independent events, and other events	11/3/26
35	Conditional probability	11/3/26
36	Bayes’ Rule	11/3/26
37	Expected value of random variable	13/3/26
38	Variance of random variable	13/3/26
39	Discrete and continuous Random Variables	17/3/26
40	Independent Random Variables	18/3/26
41	Expectation and Variance of discrete and Continuous random variable	20/3/26
42	Probability Distribution and their Properties	20/3/26
43	Binomial Distribution	31/3/26

44	Poisson distribution	31/3/26
45	Normal Distribution	1/4/26
CH 3	**Testing Of Hypothesis**	**15**
46	Population and sample, Null hypothesis and alternating hypothesis	1/4/26
47	Type-1 and Type-2 Errors	3/4/26
48	Level of significance and critical region, One-tail and two-tailed	3/4/26
49	Test of significance: Large Sample test for single proportion	7/4/26
50	Test of significance: difference of proportion	7/4/26
51	Single Mean	7/4/26
52	Difference of Means	8/4/26
53	Difference of Standard Deviations	10/4/26
54	Test for single mean (t-test)	10/4/26
55	Difference of Means (t-test)	15/4/26
56	Paired t-test	17/4/26
57	Test for Ratio of variances	21/4/26
58	Chi-square test for goodness of fit	21/4/26
59	Independence of attributes	22/4/26
60	Confidence interval	24/4/26